

PSEUDOCODE WRITING

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QUESTIONS

Q1

Student Management System
[BL3 – Apply]
10 marks

Q2

Library Book Management [BL3 – Apply]
10 marks

Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

Q8 8. E-Commerce Product Catalog [BL4 – Analyze]

10 marks

Analyze and design a Product class hierarchy for an e-commerce system:

- Electronics — with warranty, brand, voltage
- Clothing — with size, fabric, gender
- Grocery — with expiryDate and weight

Override an applyDiscount() method for each category with different discount rules. Analyze why polymorphism makes the cart checkout process flexible and extensible. OOP Concepts: Polymorphism | Inheritance | Method Overriding

Your Answer

```
class Product
    productId
    name
    basePrice
    method getPriceAfterDiscount()
        return basePrice - applyDiscount()

    method applyDiscount()
        return 0

class Electronics extends Product
```

Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9

9. University Staff Hierarchy

Your Answer

```
class Electronics extends Product
    warrantyMonths
    brand
    voltage
    method applyDiscount()
        if warrantyMonths >= 24 then
            rate = 0.15
        else
            rate = 0.08
        if brand equalsIgnoreCase "PremiumBrand" then
            rate = rate + 0.03
        return basePrice * rate

class Clothing extends Product
    size
    fabric
    gender
    method applyDiscount()
        if gender equalsIgnoreCase "Unisex" then
            rate = 0.12
        else if fabric equalsIgnoreCase "Cotton" then
            rate = 0.10
        else
            rate = 0.06
```

Q4 ✓
Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5 ✓
5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6 ✓
6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7 ✓
7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8 ✓
8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9 ✓
9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

Q10 ✓

```
else
    rate = 0.06
    return basePrice * rate

class Grocery extends Product
    expiryDate
    weightKg
    method applyDiscount()
        daysLeft = daysBetween(today, expiryDate)
        if daysLeft <= 3 then
            rate = 0.20
        else if daysLeft <= 7 then
            rate = 0.10
        else
            rate = 0.05
        return basePrice * rate

class Cart
    items = list of Product
    method checkout()
        total = 0
        for each item in items
            total = total + item.getPriceAfterDiscount()
        return total
```

📁 Save Answer